

PROJECT 3

SQL RETAIL DATA ANALYSIS REPORT

PROJECT OVERVIEW

DATABASE SETUP AND DATA PREPARATION

The raw transaction dataset initially underwent structured cleaning in **Microsoft Excel** to ensure data consistency across text and numerical fields. Once cleaned, the dataset was saved as a **CSV file** and imported directly into Microsoft SQL Server under the `DecodeLabs` database as `dbo.retail_orders`.

Key data preparation actions included:

- Cleaning missing fields and formatting standard record structures in Microsoft Excel prior to export.
- Converting the prepared Excel sheet into a standard CSV format for database ingestion.
- Ingesting the CSV file into SQL Server (`dbo.retail_orders`) and running initial schema verification queries.
- Replacing empty string entries in the `CouponCode` column with "No Coupon" to preserve aggregation accuracy.
- Validating column data types and generating time-based fields (`Order_Year` and `Order_Month`) using native SQL date extraction functions.

SQL SKILLS & TECHNIQUES APPLIED

This analysis leverages several core SQL commands and technical operations:

- **Data Querying & Filtering:** `SELECT`, `WHERE`, `ORDER BY`, `TOP`
- **Data Aggregation & Summarization:** `GROUP BY`, `HAVING`
- **SQL Aggregate Functions:** `SUM ()`, `AVG ()`, `COUNT ()`, `ROUND ()`
- **Operational Analytics:** Multi-column grouping, conditional evaluation, and channel performance modeling

KEY FINDINGS & BUSINESS METRICS

1. Overall Performance Overview

Metric Header	Quantitative Value
Total Ingested Records	1,200
Total Gross Revenue	\$1,264,761.96
Average Order Value (AOV)	\$1,053.97
Average Quantity Per Order	2 items
Total Cancelled Orders	251 orders

2. Product Revenue & Sales Volume

Product	Total Orders	Total Units Sold	Avg Unit Price (\$)	Total Revenue (\$)
Chair	178	562	355.66	195,620.11
Printer	181	542	351.71	195,612.61

Product	Total Orders	Total Units Sold	Avg Unit Price (\$)	Total Revenue (\$)
Laptop	173	535	357.71	192,126.56
Tablet	179	497	367.68	186,568.95
Monitor	163	480	358.66	175,651.41
Desk	170	508	329.61	167,459.93
Phone	156	411	375.22	151,722.39

3. Revenue by Payment Method

Payment Method	Total Transactions	Total Revenue (\$)
Credit Card	—	263,847.63
Online Payments	259	262,442.94
Cash	—	259,786.29

Payment Method	Total Transactions	Total Revenue (\$)
Gift Card	—	246,323.92
Debit Card	—	232,361.18

4. Revenue by Referral Channel

Referral Source	Total Orders	Total Revenue (\$)
Instagram	259	275,285.45
Email	250	261,808.55
Google	241	250,441.48
Facebook	228	250,410.90
Referral	222	226,815.58

CONCLUSION

This project successfully established a structured database workflow starting from initial cleaning in Excel, converting to CSV, importing into SQL Server, and running queries to extract high-value commercial insights. Key takeaways include:

1. **Top Revenue Generators:** Office hardware products (**Chairs, Printers, and Laptops**) emerged as the top three revenue contributors.
2. **Dominant Acquisition Channels:** **Instagram** and **Email** drove the highest volume of orders and overall revenue.
3. **Payment Preferences:** **Credit Cards** and **Online Payments** accounted for the largest financial volume across transaction methods.

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